

The **BLE-Based Employee Tracking System** is a modern and practical solution aimed at automating the attendance and work-hour tracking process in organizations. By leveraging **Bluetooth Low Energy (BLE)** beacons and **ESP32 microcontrollers**, the system provides a contactless, efficient, and real-time method to monitor employee entry and exit times.

The integration of **Firestore** as the backend and the implementation of a **web dashboard** ensures that data is accessible remotely and in real-time, providing enhanced transparency and ease of management for supervisors and administrators. The system reduces manual errors, eliminates the dependency on traditional attendance methods like biometric scanners or RFID cards, and saves administrative effort and time.

During the development of this project, we not only applied technical skills related to embedded systems, web development, and cloud integration but also gained valuable experience in real-world problem solving, system design, and collaborative teamwork.

This system is scalable, energy-efficient, and adaptable to different environments, from small offices to large industrial or construction sites. It has a solid foundation that can be extended in the future with features like payroll integration, mobile apps, and AI-based analytics.

In conclusion, this project demonstrates how a well-designed IoT-based system can significantly improve operational efficiency in workforce management. It aligns with the vision of smart industry solutions and serves as a stepping stone toward more intelligent and connected workplaces.